



Ansible Playbook循环语句

誉天教育

学习目标

- with_items 循环列表
- with_dict 循环字典
- 循环语句loop



- 我们在编写playbook的时候，不可避免的要执行一些重复性操作，比如指安装软件包，批量创建用户，操作某个目录下的所有文件等。正如我们所说，ansible一门简单的自动化语言，所以流程控制、循环语句这些编程语言的基本元素它同样都具备。
- 在Ansible 2.5以前，playbook通过不同的循环语句以实现不同的循环，这些语句使用with_作为前缀。这些语法目前仍然兼容，但在未来的某个时间点，会逐步废弃。

1. with_items : 循环列表

示例：批量创建组

```
- hosts: node1
gather_facts: no
vars:
  mgroup:
    - group1
    - group2
    - group3
tasks:
  - name:
    group:
      name: "{{ item }}"
      state: present
    with_items:
      "{{ mgroup }}"
```

直接在with_items中定义
被循环的列表

```
- hosts: node1
gather_facts: no
tasks:
  - name:
    group:
      name: "{{ item }}"
      state: present
    with_items:
      - group1
      - group2
      - group3
```

也可以写成如下的格式：

```
- hosts: node1
gather_facts: no
tasks:
  - name:
    group:
      name: "{{ item }}"
      state: present
    with_items:
      ["group1","group2","group3"]
```

1. with_items : 循环列表

复杂示例：批量创建用户

```
- hosts: node1
gather_facts: no
vars:
  users:
    - name: bjones
      home_dirs: /home/bjones
      uid: 2000
    - name: acook
      home_dirs: /home/acook
      uid: 2001
```

```
tasks:
  - name:
    user:
      name: "{{ item.name }}"
      home: "{{ item.home_dirs }}"
      uid: "{{ item.uid }}"
    with_items:
      "{{ users }}"
```

2. with_dict : 循环字典

- hosts: test

gather_facts: no

vars:

users:

alice:

name: Alice Appleworth

telephone: 123-456-7890

bob:

name: Bob Bananarama

telephone: 987-654-3210

tasks:

- name: Print phone records

debug:

msg: "User {{ item.key }} is {{ item.value.name }}
({{ item.value.telephone }})"

with_dict: "{{ users }}"

- 在ansible 2.5及以前的版本当中，所有的循环都是使用with_X风格。但是从2.6版本开始，官方开始推荐使用loop关键字来代替with_X风格的关键字。
- 如下示例，启动httpd和postfix服务：
 - hosts: node1
 - tasks:
 - name: postfix and httpd are running
 - service:
 - name: "{{ item }}"
 - state: started
 - loop:
 - postfix
 - httpd

- 下面是一个register的变量在循环中使用的例子：

- name: registered variable usage as a loop list

hosts: node1

tasks:

- name: ensure /mnt/bkspool exists

file:

path: /mnt/bkspool

state: directory

- name: retrieve the list of home directories

command: ls /home

register: home_dirs

- name: Show home_dirs results

debug:

var: home_dirs.stdout_lines

- name: add home dirs to the backup spooler

file:

path: /mnt/bkspool/{{ item }}

src: /home/{{ item }}

state: link

force: yes

loop: "{{ home_dirs.stdout_lines }}"

- 使用loop配合dict2items过滤器实现with_dict功能：

- hosts: test

- gather_facts: no

- vars:

- users:

- alice: female

- bob: male

- tasks:

- debug:

- msg: "{{ item.key }}" is "{{ item.value }}"

- loop: "{{users | dict2items }}"

- ✓ with_items 循环列表
- ✓ with_dict 循环字典
- ✓ 循环语句loop

